Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping **React/TypeScript** interfaces and Python and Node.js **APIs** on SQL and NoSQL databases with caching, including AI image-recognition features and marketplace payments, deployed on **AWS**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Cut dashboard latency from 1.2s to 350ms for 500+ users by redesigning **SQL** schemas, indexes, and access patterns behind a multi-tenant platform's APIs on Azure SQL and PostgreSQL.
- Built fault-tolerant FastAPI **background workers** with explicit job states and failure isolation, processing 1K+ AI jobs at 3x throughput under concurrent load while integrations held 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built full-stack features end to end, from **React** upload, history, and management views to Flask REST APIs and MySQL on AWS RDS, serving 150+ instructors and 10K+ daily requests.
- Integrated **image recognition** with GPT-4o Vision, Claude Vision, and OCR to read handwritten uploads, with concurrent processing of 1K+ uploaded files per batch and 60% faster feedback turnaround for instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Optimized high-volume APIs with indexing, API-layer **caching**, pagination, and field filtering, improving response times 30% and reducing technical debt 20% through modular object-oriented Singleton and Factory designs for low-bandwidth users.
- Wrote 20+ unit and integration **test** suites in nightly Jenkins CI/CD across three operating systems and led an AWS Glue and S3 ETL redesign boosting pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **TypeScript** and Node.js microservices with React and MongoDB for resource allocation, customer records, billing, leads, and notifications, syncing 2M+ records across multiple locations via real-time WebSockets.
- Used Google Analytics on load times and drop-off points to improve UX, cutting resource wastage 42% and raising conversions 15%, and debugged production with **Sentry** and Kibana logs.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **marketplace** APIs for a home-services booking platform covering service catalogs, time slots, pricing, coupons, and booking status with Firebase real-time updates, sustaining sub-200ms latency across 50K+ booking records.
- Integrated multi-currency **payments** with idempotent webhooks for refunds, disputes, and chargebacks, keeping payment status consistent and cutting manual payment handling by 70%, behind React TypeScript web and React Native mobile clients.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS
Frontend: React, Next.js, Angular, TailwindCSS, Responsive Design, Single-Page Applications
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Background Workers, Webhooks
Data: PostgreSQL, MySQL, MongoDB, Firebase, SQL Server, Caching, Query Optimization
Cloud & Tooling: AWS (S3, RDS, EC2), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Git, Sentry
Practices: Data Structures, Algorithms, Unit Testing, Code Reviews, Agile

Projects

Seller Listing Manager | *React, TypeScript, Node.js, PostgreSQL, Redis, AWS S3*

- Designed a seller dashboard for bulk **listing** creation with S3 photo uploads, AI-suggested titles and brands, inventory and order status views, and a Node.js API with Redis-cached **inventory** counts.

Trend Tracker with Fashion Recommendation System Live | *React.js, Material UI, Strapi, CNN, AWS EC2*

- Built a **fashion e-commerce** platform with React and Material UI on a Strapi backend and AWS EC2, adding CNN recommendations that lifted accuracy from 78% to 92% across 3M+ users.

PiXMod Live | *Next.js, MongoDB, Stripe, CLIP, SAM*

- Built a Next.js image SaaS with CLIP **visual search** over 100K+ images, SAM segmentation, MongoDB metadata storage, and Stripe subscription billing flows that raised premium conversion 18% while processing 1K+ edited assets.